OPEN

Predicting Maximal Military Occupational Task Performance from Physical Fitness Tests Using Machine Learning

AYDEN MCCARTHY[1,2], JODIE ANNE WILLS[1,2], JOEL THOMAS FULLER[1,2], STEVE CASSIDY[3], BRAD C. NINDL[4], and TIM L. A. DOYLE[1,2]

[1]Performance and Expertise Research Centre, Macquarie University, Macquarie Park, NSW, AUSTRALIA; [2]Biomechanics, Physical Performance, and Exercise Research Group, Macquarie University, Macquarie Park, NSW, AUSTRALIA; [3]School of Computing, Macquarie University, Macquarie Park, NSW, AUSTRALIA; and [4]Department of Sports Medicine and Nutrition in the School of Health and Rehabilitation Sciences, University of Pittsburgh, Neuromuscular Research Laboratory/Warrior Human Performance Research Laboratory, Pittsburgh, PA

BASIC SCIENCES

ABSTRACT

MCCARTHY, A., J. A. WILLS, J. T. FULLER, S. CASSIDY, B. C. NINDL, and T. L. A. DOYLE. Predicting Maximal Military Occupational Task Performance from Physical Fitness Tests Using Machine Learning. *Med. Sci. Sports Exerc.*, Vol. 57, No. 9, pp. 1877–1885, 2025. **Purpose:** Optimal performance in military tasks is crucial for operational success. These tasks are often simulated in training, assessing personnel performance within a military environment. However, these assessments are time-consuming and a potential injury risk. Physical characteristics such as muscular strength, power, aerobic endurance, and circumferences can be used to predict these dynamic and demanding tasks. Utilizing machine learning models to predict assessment outcomes may lead to optimized management of personnel, time, and interventions in the military. **Methods:** This study recruited 35 participants to complete two physical sessions assessing multiple physical characteristics and lift-to-place and jerry-can-carry assessments. Machine learning models were developed to predict assessment outcomes based on a down-selection of physical characteristics metrics. Root mean square error (RMSE), normalized root mean square error (NRMSE), and coefficient of variation of the root mean square error (CVRMSE) were used to evaluate the models' predictive capabilities. **Results:** The support vector regression (SVR) and ridge models could predict the lift-to-place outcome to an RMSE of ±1.77 kg (NRMSE = 4.44%, CVRMSE = 0.18) and ±2.33 kg (NRMSE = 5.84%; CVRMSE = 0.24) with four and three physical tests, respectively. The multilayer perceptron and SVR models predicted the jerry-can-carry outcome to ±3.36 laps (NRMSE = 23.06%, CVRMSE = 0.39) and ±3.67 laps (NRMSE = 25.20%, CVRMSE = 0.42) with 12 and 8 physical tests, respectively. **Conclusions:** The lift-to-place outcome can be accurately predicted, showing potential military implementation. The jerry-can-carry outcome shows promise; however, further model optimization and training metrics are required to reduce error. Machine learning models demonstrate their applicability to optimize occupational selection pathways and training interventions for desirable performance in military settings. **Key Words:** CASUALTY EVACUATION, DATA SCIENCE, OCCUPATIONAL, MANUAL MATERIEL HANDLING

Address for correspondence: Tim L. A. Doyle, Ph.D., Department of Health Sciences, Faculty of Medicine, Health, and Human Sciences, Macquarie University, 75 Talavera Rd, Macquarie Park, 2109 NSW, Australia; E-mail: tim.doyle@mq.edu.au.

Submitted for publication November 2024.
Accepted for publication April 2025.
Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.acsm-msse.org).

0195-9131/25/5709-1877/0
MEDICINE & SCIENCE IN SPORTS & EXERCISE®
Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc. on behalf of the American College of Sports Medicine. This is an open-access article distributed under the terms of the Creative Commons Attribution-Non Commercial-No Derivatives License 4.0 (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

DOI: 10.1249/MSS.0000000000003727

Performing optimally in demanding military tasks is integral to successful mission outcomes (1). Soldiers' physical characteristics are pivotal in their ability to perform tasks such as casualty evacuation, manual materiel handling, and combat maneuvers (2–6). Physical characteristics incorporate metrics such as anthropometrics, body composition, strength, power, endurance, and mobility. These metrics have been employed to predict physical assessment outcomes in military and sporting environments (1,7–11). Military organizations implement physical assessments to ensure soldiers have the physical characteristics to complete required occupational duties successfully (12–14).

Military organizations across Australia, England, Canada, and the United States have implemented nation-specific employment standards unique to their military operational requirements (12,14–16). The Australian Army incorporates a Physical Employment Standards Assessment (PESA) comprising four vital

military tasks, with minimum standards required to be achieved based on the occupational role (i.e., All Corps or Infantry). Assessments include the lift-to-place (manual materiel handling), jerry-can-carry (casualty evacuation), fire and movement (combat), and a loaded march (patrolling). Other international assessments include a casualty drag (casualty evacuation—the United States), point of entry (obstacle negotiation—the United Kingdom), and loaded intermittent shuttles (external load carriage—Canada) (14,17,18). Furthermore, international militaries generally invoke a Pre-Enlistment Fitness Assessment (PFA); a basic fitness test that must be passed before entering a military discipline. Occupational and sex-specific standards in the Australian PFA must be passed for certain physical characteristic metrics: muscular endurance (sit-ups), strength (push-ups), and cardiovascular endurance (multistage fitness test) (19). Soldiers with high levels of strength, power, and endurance generally perform better in military tasks and assessments than their peers (6,20). Obtaining simple physical characteristic metrics from PFA or physical test batteries can be utilized to develop machine learning models capable of predicting military PESA outcomes. Such machine learning models will provide access to individual candidates' predicted PESA results and insights into relevant physical characteristics. This will optimize military occupational role selections and may facilitate the development of individualized physical interventions to obtain the necessary physical characteristics for optimal military performance.

A barrier to achieving effective machine learning models is the reliability and quality of the input metric data set. Valid physical performance and characteristic metrics are vital for successful machine learning model optimization (21,22). Exercise and health tools and technological systems can measure an individual's kinetics, kinematics, body composition, and anthropometrics to profile various individuals' physical characteristics. Using validated metric measures within machine learning models can optimize training models to predict the desired outcomes of a given assessment or task to the lowest possible error (7,23). Developed models can then be cross-validated with unseen data and be assessed for prediction error and feasibility for users like the military. An accurate and feasible military machine learning model developed by Connaboy et al. (11) adopting a decision tree-like structure predicted lower limb injuries in military personnel. Although further validation of the model is required, simple and easily obtainable metrics, such as kinematic asymmetries in the knee, were identified as injury risk predictors. When assessing model feasibility, the benefits and burden of collecting the input data should not outweigh directly measuring the desired outcome.

As machine learning models are opted over traditional statistical methods, correctly selecting and applying the most appropriate model are integral to accurately predicting human movement and activity (24). Models range from linear regressions to models that can handle linear and nonlinear techniques like support vector regressions (SVR), random forests (RF), and neural networks (NN). A linear regression model like the ridge, which utilizes a regularization approach, may suffice for simple human movements and tasks that are highly correlated to physical characteristics like strength and power (25,26). Models like SVR, RF, and NN may have the suitable model architecture to handle intricate relationships and nonlinearity within advanced, dynamic, and multidimensional human movement data (7,10,27). Depending on the activity being analyzed, appropriate model selection is essential for the best prediction outcomes; thus, increasing the likelihood of model implementation in a real-world scenario.

This study aimed to develop multiple machine learning models to predict the maximal effort performance outcomes of the lift-to-place and jerry-can-carry assessments employed by the Australian Defence Force (ADF). The primary aim was to utilize simple and attainable metrics representing physical characteristics obtained via a physical test battery to develop machine learning models. The primary outcome was to explore which machine learning model was most appropriate for predicting assessment outcomes to the lowest error, providing proof of concept. In addition, it will inform the feasibility of model implementation within a military setting. It was hypothesized that at least one model's coefficient of variation would be <1 (the root mean square will be within the variability of the outcomes standard deviation) and the normalized root mean square will be less than 20% for both the lift-to-place and jerry-can-carry models developed.

METHODS

Participants

Thirty-five healthy participants (sex, 9 females and 26 males; age, 28.9 ± 8.7 yr; height, 1.73 ± 0.08 m; mass, 74.1 ± 12.3 kg) provided written, informed consent and completed this study. This research involved maximal effort requirements; hence, participants were required to be free from current mobility limitations, injuries, and medical complications at the time of collection or within 6 months before participation. Participants were screened using the Exercise Sports Science Australia Adult Pre-Exercise Screening tool (Supplemental Digital Content 1, http://links.lww.com/MSS/D223) to ensure safe participation (28). The research was approved by the Macquarie University Ethics Committee (protocol number: 520231297449904).

Experimental Protocol

An observational study was employed to evaluate the effectiveness of machine learning models to predict military assessment outcomes using physical characteristic metrics. Each participant completed two testing sessions, with at least 3-days rest between sessions. Both sessions were performed in the same indoor venue in a standardized climate. Session 1 was a physical test battery measuring physical performance and characteristic metrics (Table 1). Session 2 involved a body composition analysis, the lift-to-place and jerry-can-carry assessments, and drop landings. This study included two conditions: no body armor (NBA) and body armor (BA). Depending on the test or assessment, either one or both conditions were used, as seen

TABLE 1. The physical test battery for session 1, with tests displayed in the order that they were conducted.

Session 1 Tests	Device	Condition	Metrics
Shoulder flexion and extension	HumanTrak (VALD, Australia)	NBA and BA	Peak angles, RPE
Shoulder abduction	HumanTrak	NBA and BA	Peak angles, RPE
Overhead squat	HumanTrak	NBA and BA	Peak sagittal angles of the knee, hip, and trunk and RPE
Countermovement jump	ForceDecks (VALD, Australia)	NBA and BA	Kinetics (i.e., jump height, peak and mean forces, and power) and RPE
Squat jump	ForceDecks	NBA and BA	Kinetics (i.e., jump height, peak and mean forces, and power) and RPE
Isometric mid-thigh pull	ForceDecks	NBA	Kinetics (i.e., peak forces, and rate of force development)
Isometric grip strength	Baseline dynamometer	NBA	Kilograms
Isometric hip abduction and adduction	ForceFrame (VALD, Australia)	NBA	Kinetics (peak forces)
Isometric push-up	ForceDecks	NBA	Kinetics (i.e., peak forces, and rate of force development)
Multistage fitness test	Official audio and Polar heart rate monitor	N/A	Level, $\dot{V}O_{2max}$, peak heart rate, recovery heart rate (60 s)

Only the main metrics are listed within this table.
BA, body armor (23.5 kg); NBA, no body armor; RPE, rated perceive exertion; $\dot{V}O_{2max}$, maximal oxygen uptake per minute.

in Table 1. Standardized BA weighing 23.5 kg was donned for mobility, plyometric, and all physical tests in session 2. A BA mass of 23.5 kg has been quantified as the minimum standard load worn by ADF personnel when conducting daily military tasks (29). The BA comprised 4.5-kg front and back metal plates, 1.25-kg metal blocks, and rags distributed proportionally to minimize asymmetrical load perturbations. Supplemental Digital Content 1, http://links.lww.com/MSS/D223, documents lists all metrics obtained from sessions 1 and 2.

Upon arrival at session 1, participants' height and mass were recorded. Participants completed three repetitions of each test (the multistage fitness test was performed once) with maximal effort (Table 1). Three minutes of rest between each physical test was provided to ensure optimal physical performance. Standardized instructions and demonstrations were provided alongside verbal encouragement. Tests were ordered in a logical sequence, as displayed in Table 1, so participants would be sufficiently warmed up to reduce the likelihood of injuries. The manufacturer's protocol recommendations were followed for all HumanTrak (VALD, Queensland, Australia; V. 2.9.3), ForceDecks (VALD; V. 1.45), and ForceFrame (VALD; V. 3.4.7) protocols. The isometric push-up mirrored the protocol as per Bellar et al. (30), grip strength was measured via the Hamilton et al. (31) protocol, and the multistage fitness test protocol was completed via the established methodology from Ramsbottom et al. (32) with official test audio. The rated perceived exertion (RPE) scale protocol (33) was followed for all tests conducted under both conditions (see Table 1).

Session 2 required only mass to be recorded upon arrival. Like session 1, a fixed sequence for the tests was followed. The tests were performed in the following order: body composition, lift-to-place, drop landings, and the jerry-can-carry. Body composition analyses were completed using a 3D body scanner (Model S100X; Styku, Los Angeles, CA), adhering to the manufacturer's recommended protocols. All ADF assessments were performed until volitional failure (34). The protocols followed established methodologies: the lift-to-place, using a dumbbell as the weight lifted onto a 1.5-m platform (plyometric boxes) (29), and the jerry-can-carry assessment, which required repeated bilateral lift and carrying of 22-kg jerry cans for maximal distance (35). Figure 1 depicts the jerry-can-carry and lift-to-place assessment layout. The jerry-can-carry assessment used a standardized cadence (4.5 $km \cdot h^{-1}$) audio. The drop landing was performed at a height of 40 cm under both conditions per the manufacturer's protocol (3). Participants were provided with standardized instructions, demonstrations, and verbal encouragement. Cutoff thresholds for the ADF PESAs were put into place due to equipment and time constraints. The lift-to-place had a 57-kg maximum weight that could be lifted, and the jerry-can-carry assessment had a cutoff of 50 laps (1 lap = 25 m). The weight threshold for the lift-to-place was greater than two times the minimum standard, whereas the 50 lap

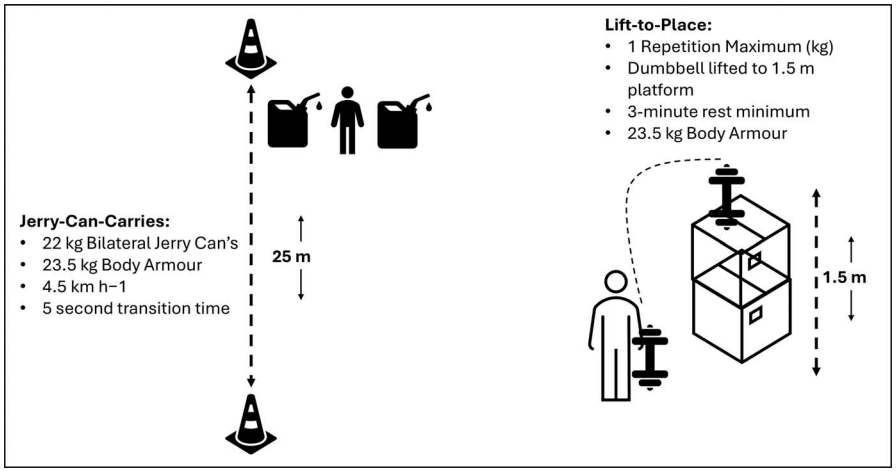


FIGURE 1—Course layout and distances for the jerry-can-carry (pictured left) and lift-to-place (pictured right) assessments.

jerry-can-carry was greater than four times the minimum standard in the ADF (34). Minimum standards for the ADF were not disclosed to encourage maximal effort. A minimum of 5 minutes of rest was given between tests and assessments. Complete assessment protocols and the ADF minimum standards can be found in Supplemental Digital Content 1, http://links.lww.com/MSS/D223.

Instrumentation, Valid Measures, and Data Processing

All kinematic data were captured using a reliable markerless 3D movement analysis system, the HumanTrak (VALD; V 2.9.3) (36,37). Data were recorded at 30 Hz and filtered using a dual-pass fourth-order Butterworth filter at 8 Hz. Custom MATLAB (R2021a; MathWorks) scripts were used to extract peak angles from the filtered data and to visually check data quality. Differences between peak angles across the conditions were also calculated. The ForceDecks (VALD; V. 1.45) and ForceFrame (VALD; V. 3.4.7) were used to capture all kinetic data. The ForceDecks sampled data at 1000 Hz; all recorded data were left unfiltered (38–40). The ForceFrame utilizes four force pads, each collecting data at 50 Hz; data also remained unfiltered (41,42). VALD's proprietary algorithms calculated all kinetic metrics (40). The participants' best trial results from each physical test were extracted from the VALD Hub (a single online centralized platform).

Body segment circumferences, surface areas, and volumes were determined using the Styku system's proprietary machine vision algorithms. The Styku utilizes a specialized camera to detect the outline of the human body, which is then used to calculate valid metrics (Macquarie University, unpublished data). Demographic information (height, sex, and mass) was also added to the participant data set.

Grip strength was measured via the baseline dynamometer (Fabrication Enterprises, White Plains, NY). The best score from the three repetitions completed was manually recorded in kilograms. The multistage fitness test was performed using the official test audio; heart rate (HR) was monitored via Polar HR monitors (H10; Polar Electro Oy, Kempele, Finland), and participants' scores and levels achieved were manually taken. In addition, peak HR, HR recovery (HR measured 60 s upon test completion), and estimated $\dot{V}O_{2max}$ were calculated and recorded (32). All PESA and RPE scores were manually recorded.

Machine Learning

Data cleaning and input metric strategies. Python's JupyterLab (V.4.0.11) was used for all data cleaning, model training and testing, and statistical analyses. A list of all Python's packages and their versions can be found in Supplementary Digital Content 1, http://links.lww.com/MSS/D223. The input metrics were separated from the outcome metric (lift-to-place and jerry-can-carry assessment scores). In addition, each test's primary metrics (Supplementary Digital Content 1, http://links.lww.com/MSS/D223) were temporarily removed from the input metrics list. The newly formed input metrics were reduced using Pearson correlation values ≥0.50 for the jerry-can-carry outcome and ≥0.75 for the lift-to-place outcome to reduce redundancy within the final input data set; these were considered secondary metrics. Correlation cutoff values were determined based on the complexity of the assessment. The jerry-can-carry was deemed a complex physical assessment, and a correlation of ≥0.50 was selected to ensure metrics that may contribute in a multivariate manner and/or a broader range of prediction variables may be required to optimize model results (43). On the other hand, the lift-to-place was deemed a simple physical assessment, and a ≥0.75 correlation would be sufficient due to the relatively single-plane nature of the movement (43). Primary metrics were then reintroduced into their respective secondary metric data set, creating the final input data set for each assessment's models. Lastly, a correlation matrix was formed with the final input data set; all pairwise correlations of ≥0.95 were identified to remove redundant metrics. The metric with a lower correlation value than the outcome metric was discarded. The outcome scores were then introduced to the final input data set and split into 80% (28 participants) training and 20% (7 participants) testing sets. A flowchart of the development of the machine learning models can be seen in Figure 2.

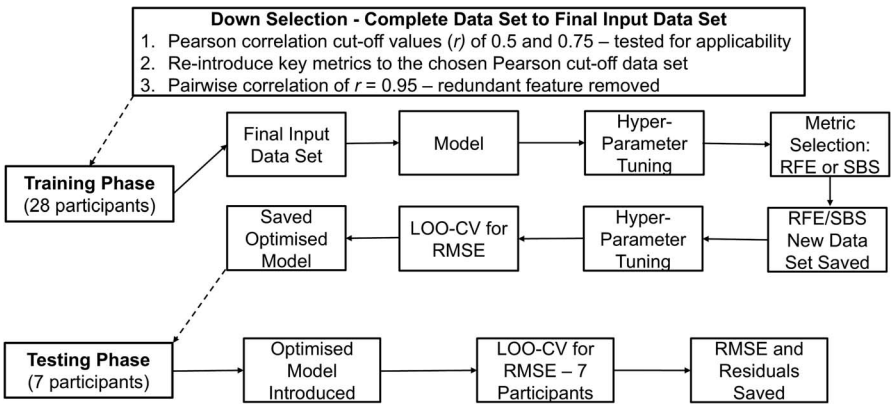


FIGURE 2—Flowchart outlining the jerry-can-carry (JCC) and lift-to-place (LP) machine learning training and testing phases for optimizing each model to reduce the root mean square error (RMSE) and the input metric set.

Training Phase

The training phase considered ridge regressions, RF, SVR, and NN (multilayer perceptron (MLP)) models to predict the lift-to-place and jerry-can-carry assessment outcomes. Models were selected based on the current literature utilizing machine learning models in an exercise and sports science setting (7,10,44–46). For all models, input metrics were standardized (*Z*-score), and a randomized grid search was utilized to optimize the model's hyperparameters. Input metrics that reduce the negative mean squared error of the model to predict the assessment outcome were then selected via a leave-one-out cross-validation using Python Scikit-learn (V. 1.4.2) recursive feature elimination (RFE) function Equation 1. A sequential backward selection (SBS) function was used for the SVR models as the RFE function is unsupported. The RFE and SBS functions' general outcome is to recursively remove metrics from the input data set until the optimal metric subset is found based on a negative mean error scoring system (47). The best metrics chosen via the RFE and SBS were then saved.

Equation 1. Recursive feature elimination with cross-validation (RFECV) function used in Python. Estimator = model used for ranking the features, step = number of features removed per iteration, cv = cross-validation − leave one out cross-validation (loo), scoring = statistical metric used to evaluate the models' performance (negative mean squared error).

$$\text{Selected_Metrics} = \text{RFECV}(\text{estimator} = \text{Model}, \text{step} = 1, \text{cv} = \text{loo}, \text{scoring} = '\text{neg_mean_squared_error}')$$

Only the optimal metric subset chosen from the RFE and SBS functions was reintroduced to each model. Due to the newly introduced optimal metric subset, a further randomized grid search was used to optimize and fit the hyperparameters for each model. All models were then assessed for their error range in predicting an individual's maximal effort for the lift-to-place and jerry-can-carry assessments using a root mean square error (RMSE). Finally, to compare the developed machine learning models against a traditional statistical approach, a forward–backward stepwise regression model (probability of F to enter = 0.05, probability of F to remove = 0.10) was also developed as a baseline for model performance comparisons.

Testing Phase

Unseen data from the testing set were tested for error accuracy with all optimized models. Error accuracy was assessed and ranked in ascending order using the lowest RMSE. Normalized RMSE (NRMSE) (Equation 2), the coefficient of variation of the RMSE (CVRMSE) (Equation 3), mean residuals, and residual standard deviation were also calculated. An NRMSE of <20% is deemed as a good fit, and a CVRMSE of <1 shows that a model's predictions are within the variability of the assessment outcome. Feasibility was described by the number of physical tests required to extract the optimal metric subset for the specified model. The demographic information collected (i.e., height, sex, and mass) was not included in the feasibility count. Each of the model scripts can be found here (48).

Equation 2. Normalized Root Mean Square Error (NRMSE) Equation.

$$\text{NRMSE} = \frac{\text{RMSE}}{\text{Mean of Outcome}} * 100$$

Equation 3. Coefficient of Variation (CV) of the the Root Mean Square Error RMSE Equation.

$$\text{CVRMSE} = \frac{\text{RMSE}}{\text{Standard Deviation of the Outcome}}$$

RESULTS

Descriptive statistics of the tests. All 35 participants completed the lift-to-place and jerry-can-carry assessments. Two participants achieved the 57-kg cutoff threshold for the lift to place, and no participants made the 50 lap jerry-can-carry cutoff. Descriptive statistics of the assessment scores are displayed in Table 2.

Machine learning model results. Table 3 conveys each model's testing phase error results in ascending order; the optimal metric count and the names and number of physical tests required for each model are also shown. The SVR model was the best developed model for the lift-to-place assessment (Table 3), although all models demonstrated an RMSE of <3.09 kg, an NRMSE of <20%, and a CVRMSE of <1. The SVR, ridge, and stepwise regression models required three identical tests, with the addition of the isometric hip abduction/adduction test being required for the SVR model (Table 3). Results for the jerry-can-carry assessment showed that all machine learning models' RMSE ranged from 3.36 to 5.31 laps and were below the <1 CVRMSE criteria; however, they were above the NRMSE threshold (20%). The stepwise regression results exhibited inferior model performance compared with the machine learning models (Table 3). The top 2 developed models for error accuracy were the MLP and SVR models, requiring 12 or 8 physical tests, respectively. Supplemental Digital Content 1, http://links.lww.com/MSS/D223, contains the training phase RMSE and hyperparameters. Supplemental Digital Content 2, http://links.lww.com/MSS/D224, contains the metrics derived by each model's RFE and SBS function.

DISCUSSION

This study evaluated four machine-learning models' ability to predict an individual's maximal effort outcome in a lift-to-place and jerry-can-carry assessment. Feasibility was also described for each model's ability to be implemented within a military setting. The hypothesis that assessment outcomes can be predicted to an acceptable error accuracy was partially

TABLE 2. Descriptive statistics of the lift-to-place and jerry-can-carry test outcomes.

Military Task Simulated Test	Min	Max	Mean	Stdev	Variance	*N*
Lift-to-place weight lifted (kg)	20.00	57.00	39.89	9.83	96.64	35.00
Jerry-can-carry laps (1 lap = 25 m)	3.00	42.00	14.57	8.69	75.44	35.00

Max, highest test score achieved of the respective test; Min, lowest test score achieved of the respective test; *N*, sample size; Stdev, standard deviation.

TABLE 3. Machine learning model testing phase results of the four models developed for the lift-to-place and jerry-can-carry assessments.

Model	RMSE (kg)	RMSE Rank	NRMSE by the Mean (%)	CVRMSE	Residual Mean	Residual Stdev	Optimal Metrics	No. of Tests	Tests
Lift-to-place									
Stepwise regression	2.84	4	7.13	0.29	–0.18	3.06	3	3	IMTP, LCMJ, Styku[a]
Ridge	2.33	2	5.84	0.24	0.20	2.18	4	3	Stkyu[a], IMTP, LCMJ, Sex[a]
Random forest	3.09	5	7.75	0.31	–0.71	3.18	15	8	IMTP, LCMJ, PU, Styku[a], CMJ, HABD, SJ, LSJ
Support vector regression	1.77	1	4.44	0.18	–0.02	1.72	7	4	Stkyu[a], IMTP, LCMJ, HABD/HADD, Sex[a]
Multilayer perceptron	2.50	3	6.27	0.25	0.61	2.50	15	8	IMTP, LCMJ, PU, Styku[a], CMJ, HABD, SJ, LSJ, Sex[a]
Jerry-can-carry									
Stepwise regression	6.14	5	42.14	0.71	–2.63	5.99	2	2	IMTP, LCMJ
Ridge	4.42	3	30.36	0.51	–1.83	4.03	70	12	Styku[a], IMTP, HABD/HADD, Grip, SJ, LSJ, CMJ, LCMJ, PU, Beep, DL, BA DL, Sex[a], Height[a]
Random forest	5.31	4	36.45	0.61	–1.33	5.14	7	4	Styku[a], LCMJ, LSJ, IMTP, Height[a]
Support vector regression	3.67	2	25.20	0.42	–0.15	3.67	15	8	IMTP, SJ, CMJ, LCMJ, Styku[a], HABD/HADD, NBA DL, BA DL, Sex[a]
Multilayer perceptron	3.36	1	23.06	0.39	1.04	3.20	37	12	Styku[a], IMTP, HABD/HADD, Grip, SJ, LSJ, CMJ, LCMJ, Height[a], PU, MSF, NBA DL, BA DL

[a] Sex and Height metrics prerequisites for the Styku system.

BA DL, body armor drop landing; CMJ, countermovement jump; CVRMSE, coefficient of variation of the root mean square error; ESSA, Exercise Science Pre-Screening Tool; Grip, grip strength; HABD/HADD, isometric hip abduction and adduction; IMTP, isometric mid-thigh pull; LCMJ, loaded countermovement jump; LSJ, loaded squat jump; MSF, multistage fitness; NBA DL, no body armor drop; No. of Tests, number of tests required; PU, isometric push-up; RHR, resting heart rate; RMSE, root mean square error; SJ, squat jump; Stdev, standard deviation; Styku, 3D body scanner.

met. All developed lift-to-place models predicted participants' lift-to-place outcome to an RMSE of ≤3.09 kg. CVRMSE and NRMSE were ≤0.31% and ≤7.75%, respectively. The SVR and ridge models were the top 2 developed models, with low error results and acceptable feasibility, as three to four physical tests were required for predictions to be made. Models developed for the jerry-can-carry assessment did not meet the hypothesis criteria, as the NRMSEs were >20%; however, the CVRMSEs were <1 for all models (Table 3). Nonetheless, the MLP and SVR models show promise with an RMSE of 3.36–3.67 laps, although 12 and 8 physical tests, respectively, are required to obtain the optimal input metrics. Results from our study suggest that the developed lift-to-place models have great potential for military implementation with low prediction error outcomes and minimal burden to collect required input metrics. On the contrary, further model optimization is required to predict jerry-can-carry outcomes due to >20% NRMSEs, potential model overfitting, and a burden on time when collecting input metrics.

To the author's knowledge, this is the first research to demonstrate effective model predictions for a military-specific lift-to-place assessment. The SVR had the lowest prediction error (RMSE = ±1.77 kg), providing proof of concept for model development and potential for military implementation. Despite the SVR model's superior predictive performance, the ridge and stepwise regression models offer a pragmatic option for military implementation, as they produce an acceptable prediction error (RMSE ±2.33–2.84 kg) with three physical tests and also align with optimal sample size-to-predictor ratios (49). The stepwise regression results reveal that traditional statistical models may still be effective when predicting simple physical assessments like the lift-to-place. The RF and MLP models also yielded acceptable prediction outcomes; however, they required a high number of predictors relative to the small sample sizes (Table 3). This increases model complexity and resource demands and reflects uncertainty about the stability of the optimal predictor variables for future investigations of the model (49). Optimal input metrics were consistent with other research (16,50–52) that indicates lower body strength and power are associated with lift-to-place tasks. Bicep circumference also contributes to the model's predictive power, possibly due to a strong correlation with the bench press and squat, similar movement patterns and muscle activations that are required throughout the lift-to-place (52,53). Although the current study did not include proven fat-free and muscular endurance metrics (16,51,54), models from similar work yielded worse performance (standard error of 13.6%) (51) compared with the current study. This study used metrics like those used in the aforementioned studies, such as lean muscle mass and isometric push-up metrics, but they did not reduce the error of our models.

Accurately predicting the lift-to-place outcome can determine an individual's manual materiel handling ability and provide insight into whether military personnel will pass the ADF PESA minimum standards (34,52). The SVR and ridge models demonstrate optimal predictive accuracy (RMSE <2.40 kg) of the weight an individual can lift, with minimal burden to complete the required physical tests, as described above. The lift-to-place incremental weight standards are 25 kg (All Corps), 30 kg (Combat Arms), 35 kg (Level 3), and 40 kg (Infantry and Special Forces recruits) (34). These standards demonstrate the demands of the different occupational role expectations and the appropriate job-specific requirements. Typically, in advanced combat roles, the volume of manual materiel handling work increases due to deployment, required training, and active combat scenarios (16). Manual materiel handling involves performing several military-specific functional movements in addition to lifting, like carrying, dragging, rope climbs, and obstacle negotiation (55,56). Special Forces recruits, for example, experience pronounced physical demands, such as heavy equipment handling during prolonged operations (6). Lift-to-place models can optimize time management, occupation role selection, and physical training interventions due to the insights into how much weight an individual can lift and highlight if they lack

specific physical characteristics. This knowledge may give information about how a recruit/soldier may perform in military environments.

The jerry-can-carry assessment is a dynamic task that encompasses multiple fitness domains compared with the lift-to-place task, which may explain the increased NRMSE ranges and a greater number of physical tests in the final model outputs compared with the lift-to-place models (Table 3). The jerry-can-carry models had a CVRMSE of <1, exhibiting that each model's prediction is within the variability range of the true assessment outcome. However, the overall model fit is outside the acceptable NMRSE range (<20%) when considering the residuals relative to the mean. Due to the multidimensional nature of the assessment, high variability is expected in the outcome. Considering one lap equals 25 m, these findings are understandable due to a standard deviation of 8.69 laps and a range of 39 laps (Table 1). For this reason, complex models, such as the NN MLP and SVR, have greater predictive capability for these more complex physical movement tasks compared with simpler models like ridge and stepwise regressions, as seen in Table 3, and as demonstrated throughout the literature (27,45,46). Interestingly, the RF model is complex, but larger datasets are typically required than our training sample of 28 to optimize results and to receive outputs like the MLP and SVR (57). It was observed that the ridge model required 70 input metrics to extract a linear relationship to predict the complex physical assessment, a notably larger number than the other models (Supplemental Digital Content 1, http://links.lww.com/MSS/D223).

The jerry-can-carry ability is predictive of lift-and-carry tasks and is often the movement pattern seen during a casualty evacuation, a demanding task required of combat soldiers (35). Due to the high physical demands of lifting and carrying, multiple physical characteristics are expressed throughout the jerry-can-carry assessment. The MLP and SVR models required 12 and 8 physical tests, respectively, to predict the assessment outcome—demonstrating the burden of completing the physical test battery and further supporting the assessments' complexity compared with the lift-to-place where only four or fewer tests were required for top-performing models (Table 3). Research demonstrates that forearm and thigh muscle cross-sectional area, push-up, grip strength, loaded $\dot{V}O_{2max}$, peak force of isometric mid-thigh pull, and body composition metrics relate to lift-and-carry and casualty evacuation task performance (16,58,59). These metrics are like the physical tests predicted by our RFE and SBS algorithms; however, lower body power and drop landing metrics were seen in the MLP and SVR models' outputs. This supports the dynamic nature of the task and provides insights into physical characteristic exercise interventions to enhance assessment results. Intriguingly, grip strength, a proven predictor metric of lift-and-carry tasks, was only found in the ridge and MLP models (35). Although our results show promise for future implementation in military settings, the MLP and SVR models require further data and model optimization to reduce RMSEs and improve sample size- and outcome-to-predictor ratios (49). Currently, the extensive physical test batteries required to obtain the input metric datasets are burdensome. Furthermore, the small sample size may lead to overfitting during model development. Model development with additional military-specific population datasets will increase the sample size and allow for greater generalizability and optimal hyperparameter selection, and may reduce the number of predictor metrics required for successful model performance (57). Future work could also consider using classifier rather than regressor modeling if unsatisfactory model errors persist despite increasing sample size.

Models developed for the lift-to-place and jerry-can-carry assessments can provide knowledge of an individual's capacity for highly demanding military assessments and tasks. Developed models must satisfy prediction error criteria; however, it is equally important to evaluate the feasibility of the data collection process required for input metrics (60). A balance between model performance and the effort associated with data collection processes is critical for facilitating model implementation in the military. The lift-to-place SVR and ridge models satisfy both low prediction error results and military implementation feasibility. The physical tests extracted from each model results can also provide insight into physical characteristics such as lower body strength and power from the IMTP and LCMJ, and estimate upper body strength from an individual's bicep circumference (53,61). Admittedly, there is an arguable case for all developed lift-to-place models; upon model validation, the end-user can select a model based on its prediction error and the physical information acquired from the input metrics (Table 3). The jerry-can-carry models require further model refinement, and currently, up to 12 physical tests are needed to develop the models (Table 3). Although models like the MLP and SVR show promise in predicting the outcome of the jerry-can-carry, further data and model optimization are required to reduce error accuracy and necessary physical tests. Subsequent model iterations should seek to determine highly influential input metrics from physical tests to streamline testing protocols and optimize model prediction outcomes. Overall, our findings suggest that military organizations might benefit from the lift-to-place models and their optimal metric inputs to adapt current physical test batteries for pre-enlistment and active-duty soldiers. Models and metrics can provide lift-to-place assessment scores and intervention and training strategies for personnel lacking in manual materiel handling efficiency. Future research can investigate a complete test battery that can predict multiple military task outcomes, minimizing resources and capturing the interconnected physical nature of multiple military tasks.

This study demonstrates the potential use of machine learning models in a military setting; however, some limitations were documented. The lift-to-place assessment had two participants meet the threshold of 57 kg. This can bias our developed models by underestimating maximal lift performance and misrepresenting metric-outcome relationships. This misrepresentation might lead to inaccurate lift-to-place predictions for individuals with high strength levels and reduced generalizability to the upper range of prediction outcomes. Another limitation

was that the population used in this study was not military personnel but the general population, who have not been exposed to military training or the environment. Military personnel often exhibit elevated strength, specific to military tasks compared with the general population, and model predictions may not accurately reflect the assessment outcomes of military populations. Future research requires cross-validation of prediction outcomes, specifically the lift-to-place models. Our study had a sample size of 35; subsequent research should focus on investigating the influence of an increased sample size on model performance, as large data sets may capture increased individual variance of physical characteristics and performance, leading to improved predictive power, model fit, and generalizability (62). Furthermore, our input metric approach used Pearson correlation cutoff values and RFE and SBS algorithms to reduce the training dataset size. Future model development approaches may reconsider this approach and follow guidelines for predetermined observations-to-predictors ratios or utilize different data dimension reduction techniques (47,63).

CONCLUSIONS

Machine learning models allow organizations like the military to enhance their objective decision-making on personnel selection and physical development. The SVR and ridge models displayed great potential use as a tool to predict the lift-to-place assessment. These models could predict the weight lifted to a 1.5-m platform to <2.40 kg with minimal physical tests required to be completed. The MLP and SVR models showed promise to predict jerry-can-carry distance with an RMSE of 3.36 and 3.67 laps, respectively, although further model optimization to reduce the error accuracy and increase feasibility is required. With further validation, our models' potential utility in estimating military assessment outcomes can lead to targeted military training interventions and help refine new and existing military personnel to specific occupational roles based on their physical characteristics and occupational ability in manual materiel handling and casualty evacuation tasks. Machine learning model development shows promise in improving military selection pathways and developing training interventions and strategies for desirable military traits.

This research was partially supported by the Macquarie University Research Excellence Stipend Scholarship, VALD Performance Stipend Scholarship under Grant 20224931, and Office of Naval Research Grant under Grant N62909-21-1-2015. This project received partial funding from VALD and the Office of Naval Research Global; they did not influence the findings or reporting of results. The authors thank all participants for volunteering in this research study. Additional acknowledgments go to Isabella Rubino and David Ahn for their data collection and set-up assistance; VALD for the in-kind supply of the HumanTrak, ForceDecks, and ForceFrame and their help and support; and Dr. Aaron Beach for his support in the methodology, set-up, and analysis of results. The results of this study are presented clearly and honestly, without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute an endorsement by the American College of Sports Medicine. The raw data supporting this article's conclusions may be made available by the authors upon a valid request to the corresponding author due to the sensitivity of the data.

REFERENCES

1. Pihlainen K, Santtila M, Hakkinen K, Kyrolainen H. Associations of physical fitness and body composition characteristics with simulated military task performance. *J Strength Cond Res*. 2018;32(4):1089–98.
2. Botta WC, Franchini E, Gabriel-Costa D, Campos FAD. Physical tests to predict combat task performance among Brazilian Air Force Infantry cadets. *Mil Med*. 2023;188(9–10):3095–101.
3. Doyle TLA, Fain AC, Wills JA, Cooper D, Toonen K, Kamphius B. Measures of lower body strength associated with injuries in Australian Special Forces selection candidates. *J Appl Biomech*. 2022;38(4): 255–62.
4. Hauschild VD, DeGroot DW, Hall SM, et al. Fitness tests and occupational tasks of military interest: a systematic review of correlations. *Occup Environ Med*. 2017;74(2):144–53.
5. Heinrich KM, Spencer V, Fehl N, Poston WS. Mission essential fitness: comparison of functional circuit training to traditional army physical training for active duty military. *Mil Med*. 2012;177(10):1125–30.
6. Gagnon P. *Section 7.11—Effects of Extended Operations on Physical Performance*. Brussels (Belgium): NATO Research and Technology Organisation; 2008. pp. 7-56–61 Available from: NATO Research and Technology Organisation.
7. Ahamed NU, Krajewski KT, Johnson CC, et al. Using machine learning and wearable inertial sensor data for the classification of fractal gait patterns in women and men during load carriage. *Procedia Comput Sci*. 2021;185:282–91.
8. Allison KF, Keenan KA, Lovalekar M, et al. Fight load index and body composition are most associated with combat fitness in female marines. *J Sci Med Sport*. 2019;22(4):494–9.
9. Bird MB, Mi Q, Koltun KJ, et al. Unsupervised clustering techniques identify movement strategies in the countermovement jump associated with musculoskeletal injury risk during US Marine Corps Officer Candidates School. *Front Physiol*. 2022;13:868002.
10. Bongiovanni T, Trecroci A, Cavaggioni L, et al. Importance of anthropometric features to predict physical performance in elite youth soccer: a machine learning approach. *Res Sports Med*. 2021;29(3): 213–24.
11. Connaboy C, Eagle SR, Johnson C, Flanagan S, Mi Q, Nindl BC. Employing machine learning to predict lower extremity injury in US Special Forces. *Med Sci Sports Exerc*. 2019;50(5):1073–709.
12. Abdel-messih M. Australian Army physical training continuum-reducing injury and medical discharge rates. *Aust Army J*. 2023;19(1):37–54.
13. Department of Defence. (n.d.). Physical employment standards. Defence Science and Technology Group. Available from: https://www.dst.defence.gov.au/projects/physical-employment-standards.
14. Treweek AJ, Tipton MJ, Milligan GS. Development of a physical employment standard for a branch of the UK military. *Ergonomics*. 2019;62(12):1572–84.
15. Sharp M. Pre-enlistment physical employment standards screening of US Army recruits. *J Sci Med Sport*. 2017;20:S8–9.
16. Jaenen S. Identification of common military tasks. *Optim Operat Phys Fitness*. 2009;8:1–6.
17. Foulis SA, Sharp MA, Redmond JE, et al. U.S. Army physical demands study: development of the occupational physical assessment test for combat arms soldiers. *J Sci Med Sport*. 2017;20(Suppl 4):S74–8.
18. Chassé E, Laroche M-A, Dufour C-A, Guimond R, Lalonde F. Association between musculoskeletal injuries and the Canadian Armed Forces physical employment standard proxy in Canadian military recruits. *Mil Med*. 2020;185(7–8):e1140–6.
19. Australian Defence Force [Internet]. ADF. Available from: https://www.adfcareers.gov.au/joining/can-i-join/health-and-fitness.
20. Vaara JP, Groeller H, Drain J, et al. Physical training considerations for optimizing performance in essential military tasks. *Eur J Sport Sci*. 2022;22(1):43–57.

21. Hughes GT, Camomilla V, Vanwanseele B, Harrison AJ, Fong DT, Bradshaw EJ. Novel technology in sports biomechanics: some words of caution. *Sports Biomech*. 2024;23(4):393–401.
22. Gupta N, Mujumdar S, Patel H, et al. Data quality for machine learning tasks. In: *Proceedings of the Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*. Singapore. pp. 4040–1.
23. Rommers N, Rossler R, Verhagen E, et al. A machine learning approach to assess injury risk in elite youth football players. *Med Sci Sports Exerc*. 2020;52(8):1745–51.
24. Horvat T, Job J. The use of machine learning in sport outcome prediction: a review. *WIRES Data Min Knowl*. 2020;10(5):e1380.
25. Sandau I, Kipp K. Prediction of snatch and clean and jerk performance from physical performance measures in elite male weightlifters. *J Strength Cond Res*. 2025;39(1):33–40.
26. Bishop CM, Nasrabadi NM. *Pattern Recognition and Machine Learning*. Cham (Switserland): Springer; 2006.
27. Maszczyk A, Zajac A, Rygula I. A neural network model approach to athlete selection. *Sports Eng*. 2011;13(2):83–93.
28. Norton K, Norton L. Pre-exercise Screening: Guide to the Australian adult pre-exercise screening system. Canberra: Exercise and Sports Science Australia/Fitness Australia/Sports Medicine Australia; 2011. p. 53.
29. Carstairs GL, Ham DJ, Savage RJ, Best SA, Beck B, Doyle TL. A box lift and place assessment is related to performance of several military manual handling tasks. *Mil Med*. 2016;181(3):258–64.
30. Bellar D, Marcus L, Judge LW. Validation and reliability of a novel test of upper body isometric strength. *J Hum Kinet*. 2015;47(1):189–95.
31. Hamilton A, Balnave R, Adams R. Grip strength testing reliability. *J Hand Ther*. 1994;7(3):163–70.
32. Ramsbottom R, Brewer J, Williams C. A progressive shuttle run test to estimate maximal oxygen uptake. *Br J Sports Med*. 1988;22(4):141–4.
33. Borg G. Perceived exertion as an indicator of somatic stress. *Scand J Rehabil Med*. 1970;2(2):92–8.
34. Australian Defence Force [Internet]. Available from: https://content.defencejobs.gov.au/pdf/army/Army_Physical_Continuum_Information.pdf.
35. Beck B, Middleton KJ, Carstairs GL, Billing DC, Caldwell JN. Predicting stretcher carriage: investigating variations in bilateral carry tests. *Appl Ergon*. 2016;55:124–32.
36. McCarthy A, Wills JA, Lenton GK, Doyle TLA. Evaluating the intra- and inter-day reliability of output measures for the VALD HumanTrak: dynamic movements and range of motion of the shoulder and hip with body armour. *Ergonomics*. 2023;66(3):406–18.
37. McCarthy A, Wills JA, Fain A, Nindl BC, Beach AJ, Doyle TLA. *VALD HumanTrak Measures Valid Hip and Knee Joint Angles with and without Body Armour*. Macquarie Park (Australia): Macquarie University; 2024.
38. Dos' Santos T, Lake J, Jones PA, Comfort P. Effect of low-pass filtering on isometric midthigh pull kinetics. *J Strength Cond Res*. 2018; 32(4):983–9.
39. Street G, McMillan S, Board W, Rasmussen M, Heneghan JM. Sources of error in determining countermovement jump height with the impulse method. *J Appl Biomech*. 2001;17(1):43–54.
40. Collings TJ, Lima YL, Dutaillis B, Bourne MN. Concurrent validity and test–retest reliability of VALD ForceDecks strength, balance, and movement assessment tests. *J Sci Med Sport*. 2024;27(8):572–80.
41. Ryan S, Kempton T, Pacecca E, Coutts AJ. Measurement properties of an adductor strength-assessment system in professional Australian footballers. *Int J Sports Physiol Perform*. 2019;14(2):256–9.
42. O'Connor C, McIntyre M, Delahunt E, Thorborg K. Reliability and validity of common hip adduction strength measures: the ForceFrame strength testing system versus the sphygmomanometer. *Phys Ther Sport*. 2023;59:162–7.
43. Liu H, Motoda H. *Feature Selection for Knowledge Discovery and Data Mining*. Berlin/Heidelberg (Germany): Springer Science & Business Media; 2012.
44. Masum AKM, Hossain ME, Humayra A, Islam S, Barua A, Alam GR. A statistical and deep learning approach for human activity recognition. In: *Proceedings of the 2019 3rd International Conference on Trends in Electronics and Informatics (ICOEI)*. 2019. pp. 1332–7.
45. Przednowek K, Iskra J, Maszczyk A, Nawrocka M. Regression shrinkage and neural models in predicting the results of 400-metres hurdles races. *Biol Sport*. 2016;33(4):415–21.
46. Maszczyk A, Roczniok R, Waskiewicz Z, et al. Application of regression and neural models to predict competitive swimming performance. *Percept Mot Skills*. 2012;114(2):610–26.
47. Kuhn M, Johnson K. *Feature Engineering and Selection: A Practical Approach for Predictive Models*. Boca Raton (FL): Chapman and Hall/CRC; 2019.
48. McCarthy A Zenodo [Internet]. Available from: https://doi.org/10.5281/zenodo.14043036.
49. Hua J, Xiong Z, Lowey J, Suh E, Dougherty ER. Optimal number of features as a function of sample size for various classification rules. *Bioinformatics*. 2005;21(8):1509–15.
50. Rayson M, Holliman D, Belyavin A. Development of physical selection procedures for the British Army. Phase 2: relationship between physical performance tests and criterion tasks. *Ergonomics*. 2000;43(1):73–105.
51. Beckett MB, Hodgdon JA. *Lifting and Carrying Capacities Relative to Physical Fitness Measures*. San Diego (CA): Naval Health Research Center; 1987.
52. Savage RJ, Best SA, Carstairs GL, Ham DJ, Doyle TL. On the relationship between discrete and repetitive lifting performance in military tasks. *J Strength Cond Res*. 2014;28(3):767–73.
53. Keogh J, Hume P, Mellow P, Pearson S. The use of anthropometric variables to predict bench press and squat strength in well-trained strength athletes. In: 23rd Proceedings of the International Society of Biomechanics in Sports. 2005.
54. Hodgdon JA, & Beckett MB. Body composition and physical performance: estimation of lifting and carrying from fat-free mass (Report No. 98-37). Naval Health Research Center: 1998. Available from: https://apps.dtic.mil/sti/pdfs/ADA370123.pdf.
55. Sharp M, Rosenberger M, & Knapik J. Chapter 5 - Common military task: materials handling (Report No. MISC. 06-31). U.S. Army Research Institute of Environmental Medicine: 2006. USA.
56. Feigel ED, Sterczala AJ, Krajewski KT, et al. Physiological characteristics predictive of passing military physical employment standard tasks for ground close combat occupations in men and women. *Eur J Sport Sci*. 2024;24(9):1247–59.
57. James G, Witten D, Hastie T, Tibshirani R. *An Introduction to Statistical Learning*. New York, NY: Springer; 2013.
58. Chasse E, Tingelstad HC, Needham-Beck SC, Reilly T. Factors affecting performance on an army urban operation casualty evacuation for male and female soldiers. *Mil Med*. 2019;184(11–12): e856–62.
59. Knapik JJ, Harper W, Crowell HP. Physiological factors in stretcher carriage performance. *Eur J Appl Physiol*. 1999;79(5):409–13.
60. Larsson J, Dencker M, Olsson MC, Bremander A. Development and application of a questionnaire to self-rate physical work demands for ground combat soldiers. *Appl Ergon*. 2020;83:103002.
61. Johnson QR, Dawes JJ, Uftring M, et al. Differences in stronger versus weaker firefighters in selected measures of power. *Int J Excerc Sci*. 2022;15(4):552–60.
62. Tyagi AK. Machine learning with big data. In: *Proceedings of the Machine Learning with Big Data (March 20, 2019). Proceedings of International Conference on Sustainable Computing in Science, Technology and Management (SUSCOM)*. Jaipur-India: Amity University Rajasthan. p. 2019.
63. Dobbin KK, Simon RM. Sample size planning for developing classifiers using high-dimensional DNA microarray data. *Biostatistics*. 2007;8(1):101–17.